

BIOLOGICAL BOSE CONDENSATION AND THE TIME THRESHOLD FOR BIOLOGICAL EFFECTS[☆]

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Fröhlich's model of Bose condensation in biological systems is analyzed using finite temperature Green's function techniques. The appropriate biological system excitation lifetime is calculated and represents the time needed to produce the condensed state. The possible relevance of this calculated lifetime to experiments concerning the microwave irradiation of biological organisms is discussed.

The dipole-dipole interaction within biological structure first suggested by Fröhlich [1-6] has stimulated many recent theoretical studies [7-12, 19-21]. Also some recent experimental reports [13-18] have appeared to conform in several aspects to the predictions of the Fröhlich model. The purpose of this paper is to extend the theoretical analysis utilizing a Green's function technique and to discuss the results in relation to biological experimentation that has been performed up to the present.

Fröhlich's model has been particularly useful in understanding recent experiments revealing the effects of low level microwaves on various biological systems. One important general characteristic that is reported in these experiments is the existence of a time threshold for the initiation of the differing biological effects. Sevastyanova and Vilenskaya [13] performed an experiment concerning the simultaneous irradiation of mice with X-rays and microwaves where they counted surviving bone marrow cells. The time threshold they determined before a protective effect occurred with simultaneous irradiation was 30 minutes of microwave exposure. Smolyanskaya and Vilenskaya [18] observed the effects of low level millimeter waves on the induction coefficient of colicin synthesis. Again, it was

found that a minimum irradiation time is needed to produce measurable biological effects. The time threshold for this experiment was of the order of 30 minutes. One must keep in mind when considering these time effects that the threshold will depend on the particular biological activity being monitored and will also be temperature dependent. An example of the latter is clearly observed by analyzing Smolyanskaya and Vilenskaya's experiment. Irradiation for 30 minutes at $T = 20^{\circ}\text{C}$ produced no change in the rate of colicin synthesis, whereas at $T = 37^{\circ}\text{C}$, colicin synthesis was induced after 30 minutes of irradiation time. Therefore, it is expected that the analysis of the time threshold will not be a trivial matter. Both the detailed microscopic structure of the biological system and its thermodynamic behavior must be considered to provide a full understanding of the specific time thresholds encountered for different biosystems.

It is the authors' present intention to obtain an approximation for a segment of this time threshold using the Fröhlich model. In applying the Fröhlich hypothesis, one sees that the biological effects cannot begin until the condensed phase occurs. How long this condensation has to be maintained in order to produce biological effects is undetermined as yet. Most generally, the time needed to initiate biological effects will consist of two parts, τ_1 and τ_2 . τ_1 represents the time from the beginning of irradiation to the onset of condensation. τ_2 represents the time elapsed from the on-

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set of the condensed phase to the production of the actual biological effects. Thus, the total time, τ , elapsed before biological effects occur from the initiation of irradiation, is given by $\tau = \tau_1 + \tau_2$. In the following analysis, we calculated τ_1 to the lowest order in coupling strength using a Green's function method. An explicit form for τ_2 requires an in-depth biochemical analysis of the particular system under scrutiny and is beyond the scope of the present study. Undoubtedly, τ_2 represents an important part and should be studied in the future.

The hamiltonian of the biological system is:

$$\begin{aligned}
 H = & \sum_p \omega_p a_p^\dagger a_p + \sum_{p'} \Omega_{p'} b_{p'}^\dagger b_{p'} + \sum_q \bar{\Omega}_q p_q^\dagger p_q \\
 & + \sum_{p,p',p''} (\lambda a_p b_{p'}^\dagger b_{p''} + \lambda^* a_{p'}^\dagger b_p b_{p''}^\dagger) \\
 & + \frac{1}{2} \sum_{p,p',p''} (\alpha a_p b_{p'}^\dagger b_{p''}^\dagger + \alpha^* a_{p'}^\dagger b_p b_{p''}) \\
 & + \sum_{p,p',p''} (\chi a_p a_{p'}^\dagger b_{p''}^\dagger + \chi^* a_{p'}^\dagger a_p b_{p''}) \\
 & + \frac{1}{2} \sum_{p,p',p''} (\beta a_p a_{p'} b_{p''}^\dagger + \beta^* a_{p'}^\dagger a_p b_{p''}) \\
 & + \sum_{p,q} (\xi p_q a_p^\dagger + \xi^* p_q^\dagger a_p). \quad (1)
 \end{aligned}$$

All notations and the meanings of the various coupling constants and operators are the same as those given in ref. [8]. For simplicity in the calculation, the excitations will be regarded as phonons for both the biological system and the heat bath. However, the heat bath quanta may also consist of fermions [8]. It should be noted that the lifetime (and thus self-energy) of interest is due to the terms with coupling constants χ and χ^* because these terms are responsible for creating the condensed phase. The contributions from the other terms that involve the simultaneous creation or destruction of two quanta within the biological system (i.e., those terms with the β and β^* coupling constants) are negligible compared to the above as shown in ref. [8]. The rest of the terms in the hamiltonian do not contribute to the condensation to the lowest order in coupling constant and therefore are not relevant to the present calculation.

Using finite temperature Green's function techniques the self-energy, $\Sigma(p, \omega_p)$, for the biological system phonon associated with operators a_p and $a_p^\dagger$ was calculated yielding:

$$\begin{aligned}
 \Sigma(p, \omega_p) = & |\chi|^2 \sum_{p'} \left[\frac{1 + n(\Omega_{p'}) + n(\omega_{p-p'})}{\omega_p - \Omega_{p'} - \omega_{p-p'} + i\delta} \right. \\
 & \left. + \frac{n(\Omega_{p'}) - n(\omega_{p-p'})}{\omega_p + \Omega_{p'} - \omega_{p-p'} + i\delta} \right], \quad (2)
 \end{aligned}$$

where $n(\omega) = [\exp(\omega/kT) - 1]^{-1}$, Ω_p is the energy of a phonon with momentum p in the heat bath and ω_p is the energy of a phonon with momentum p in the biological system. Since at room temperature, these energies are much smaller than kT , $n(\omega_p)$ and $n(\Omega_{p'})$ are much larger than 1. Also $\Omega_{p'} = \pm(\omega_p - \omega_{p-p'})$ which is smaller than the width of the narrow energy band of the biological system under consideration which in turn is smaller than a typical energy in the set of vibrational modes (i.e., $\Omega_{p'} \ll \omega_p$). Thus, $n(\Omega_{p'}) \gg n(\omega_{p-p'})$. Therefore, the self-energy can be approximately written as:

$$\begin{aligned}
 \Sigma(p, \omega_p) = & \frac{|\chi|^2 kT}{4\pi^2 \hbar^3} \int_{-1}^1 dx \int_0^\infty \frac{p'^2 dp'}{\Omega_{p'}} \\
 & \times \left[\frac{1}{\omega_p - \Omega_{p'} - \omega_{p-p'} + i\delta} \right. \\
 & \left. + \frac{1}{\omega_p + \Omega_{p'} - \omega_{p-p'} + i\delta} \right], \quad (3)
 \end{aligned}$$

where the sum over heat bath momenta has been converted to an integration. Using the Debye model, for the excitations in the heat bath, $\Omega_{p'} = v'p'$, where v' is the velocity of sound in the heat bath. The form of the dispersion relation for the biological system is unknown and depends in an intricate manner upon the biological structure. To obtain an approximate result, however, it is presumed that $\omega_p - \omega_{p-p'} = \pm\gamma\Delta$ where Δ is the width of the energy band (which is assumed narrow) and γ is a positive quantity less than 1. The value of γ depends on the form of the dispersion relation for the biological system.

Using the relation:

$$\tau_1 = \hbar / |\text{Im } \Sigma|, \quad (4)$$

and the imaginary part of the self-energy in eq. (3):

$$\text{Im } \Sigma \doteq -|\chi|^2 kT\gamma\Delta/\pi v'^3 \hbar^3, \quad (5)$$

we have:

$$\tau_1 \doteq \hbar^4 \pi v'^3 / |\chi|^2 kT\gamma\Delta, \quad (6)$$

which is inversely proportional to T and to the square of the coupling constant. As seen from eq. (6), τ_1 decreases with increasing temperature. Thus, as the temperature is raised, the condensed phase should occur sooner also causing the biological effects to begin sooner. This agrees, at least qualitatively, with the example stated previously from Smolyanskaya and Vilenskaya's experiment where changing T from 20°C to 37°C brought the biological effects into play holding the irradiation time fixed at 30 minutes.

τ_1 given in eq. (6) represents the irradiation time needed to produce the condensation and is the minimum time needed to produce biological effects. In order to apply this equation to specific circumstances, model studies and more experiments need to be performed to obtain expressions for the coupling constants and the widths of the energy bands. Given this information, a more complete understanding of the Fröhlich model and all its implications can be obtained.

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